

1. One vertex of a rectangular parallelepiped is at the origin O and the lengths of its edges along x , y and z axes are 3, 4 and 5 units respectively. Let P be the vertex $(3, 4, 5)$. Then the shortest distance between the diagonal OP and an edge parallel to z axis, not passing through O or P is

[2023 (06 Apr Shift 1)]

- (1) $\frac{12}{\sqrt{5}}$
(2) $12\sqrt{5}$
(3) $\frac{12}{5\sqrt{5}}$
(4) $\frac{12}{5}$

2. If the equation of the plane passing through the line of intersection of the planes $2x - y + z = 3$, $4x - 3y + 5z + 9 = 0$ and parallel to the line $\frac{x+1}{-2} = \frac{y+3}{4} = \frac{z-2}{5}$ is $ax + by + cz + 6 = 0$, then $a + b + c$ is equal to

[2023 (06 Apr Shift 1)]

- (1) 12
(2) 14
(3) 16
(4) 13

3. Let the image of the point $P(1, 2, 3)$ in the plane $2x - y + z = 9$ be Q . If the coordinates of the point R are $(6, 10, 7)$, then the square of the area of the triangle PQR is _____.

[2023 (06 Apr Shift 1)]

4. Let the line L pass through the point $(0, 1, 2)$, intersect the line $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and be parallel to the plane $2x + y - 3z = 4$. Then the distance of the point $P(1, -9, 2)$ from the line L is

[2023 (06 Apr Shift 2)]

- (1) $\sqrt{74}$
(2) $\sqrt{69}$
(3) $\sqrt{54}$
(4) 9

5. A plane P contains the line of intersection of the plane $\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = 6$ and $\vec{r} \cdot (2\hat{i} + 3\hat{j} + 4\hat{k}) = -5$. If P passes through the point $(0, 2, -2)$, then the square of distance of the point $(12, 12, 18)$ from the plane P is

[2023 (06 Apr Shift 2)]

- (1) 620
(2) 155
(3) 310
(4) 1240

6. If the lines $\frac{x-1}{2} = \frac{2-y}{-3} = \frac{z-3}{\alpha}$ and $\frac{x-4}{5} = \frac{y-1}{2} = \frac{z}{\beta}$ intersect, then the magnitude of the minimum value of $8\alpha\beta$ is _____.

[2023 (06 Apr Shift 2)]

7. The shortest distance between the lines $\frac{x-4}{4} = \frac{y+2}{5} = \frac{z+3}{3}$ and $\frac{x-1}{3} = \frac{y-3}{4} = \frac{z-4}{2}$ is

[2023 (08 Apr Shift 1)]

- (1) $6\sqrt{3}$
(2) $2\sqrt{6}$
(3) $6\sqrt{2}$
(4) $3\sqrt{6}$

8. If the equation of the plane containing the line $x + 2y + 3z - 4 = 0 = 2x + y - z + 5$ and perpendicular to the plane

$\vec{r} = (\hat{i} - \hat{j}) + \lambda(\hat{i} + \hat{j} + \hat{k}) + \mu(\hat{i} - 2\hat{j} + 3\hat{k})$ is $ax + by + cz + 4 = 0$ then $(a - b + c)$ is equal to

[2023 (08 Apr Shift 1)]

- (1) 18
(2) 22
(3) 20
(4) 24

9. Let λ_1, λ_2 be the values of λ for which the points $(\frac{5}{2}, 1, \lambda)$ and $(-2, 0, 1)$ are at equal distance from the plane $2x + 3y - 6z + 7 = 0$. If $\lambda_1 > \lambda_2$ then the distance of the point $(\lambda_1 - \lambda_2, \lambda_2, \lambda_1)$ from the line $\frac{x-5}{1} = \frac{y-1}{2} = \frac{z+7}{2}$ is _____

[2023 (08 Apr Shift 1)]

10. For $a, b \in \mathbb{Z}$ and $|a - b| \leq 10$, let the angle between the plane $P : ax + y - z = b$ and the line $L : x - 1 = a - y = z + 1$ be $\cos^{-1}\left(\frac{1}{3}\right)$. If the distance of the point $(6, -6, 4)$ from the plane P is $3\sqrt{6}$, then $a^4 + b^2$ is equal to
[2023 (08 Apr Shift 2)]
(1) 32
(2) 85
(3) 25
(4) 48
11. Let P be the plane passing through the line $\frac{x-1}{1} = \frac{y-2}{-3} = \frac{z+5}{7}$ and the point $(2, 4, -3)$. If the image of the point $(-1, 3, 4)$ in the plane P is (α, β, γ) , then $\alpha + \beta + \gamma$ is equal to
[2023 (08 Apr Shift 2)]
(1) 10
(2) 12
(3) 9
(4) 11
12. Let P_1 be the plane $3x - y - 7z = 11$ and P_2 be the plane passing through the points $(2, -1, 0)$, $(2, 0, -1)$, and $(5, 1, 1)$. If the foot of the perpendicular drawn from the point $(7, 4, -1)$ on the line of intersection of the planes P_1 and P_2 is (α, β, γ) , then $\alpha + \beta + \gamma$ is equal to
[2023 (08 Apr Shift 2)]
13. Let O be the origin and the position vector of the point P be $-\hat{i} - 2\hat{j} + 3\hat{k}$. If the position vectors of the points A , B and C are $-2\hat{i} + \hat{j} - 3\hat{k}$, $2\hat{i} + 4\hat{j} - 2\hat{k}$ and $-4\hat{i} + 2\hat{j} - \hat{k}$ respectively, then the projection of the vector $\vec{OP}$ on a vector perpendicular to the vectors $\vec{AB}$ and $\vec{AC}$ is
[2023 (10 Apr Shift 1)]
(1) 3
(2) $\frac{8}{3}$
(3) $\frac{7}{3}$
(4) $\frac{10}{3}$
14. Let two vertices of a triangle ABC be $(2, 4, 6)$ and $(0, -2, -5)$, and its centroid be $(2, 1, -1)$. If the image of the third vertex in the plane $x + 2y + 4z = 11$ is (α, β, γ) , then $\alpha\beta + \beta\gamma + \gamma\alpha$ is equal to
[2023 (10 Apr Shift 1)]
(1) 70
(2) 76
(3) 74
(4) 72
15. The shortest distance between the lines $\frac{x+2}{1} = \frac{y}{-2} = \frac{z-5}{2}$ and $\frac{x-4}{1} = \frac{y-1}{2} = \frac{z+3}{0}$ is
[2023 (10 Apr Shift 1)]
(1) 8
(2) 6
(3) 7
(4) 9
16. Let P be the point of intersection of the line $\frac{x+3}{3} = \frac{y+2}{1} = \frac{1-z}{2}$ and the plane $x + y + z = 2$. If the distance of the point P from the plane $3x - 4y + 12z = 32$ is q , then q and $2q$ are the roots of the equation
[2023 (10 Apr Shift 1)]
(1) $x^2 - 18x - 72 = 0$
(2) $x^2 - 18x + 72 = 0$
(3) $x^2 + 18x + 72 = 0$
(4) $x^2 + 18x - 72 = 0$
17. Let the image of the point $P(1, 2, 6)$ in the plane passing through the points $A(1, 2, 0)$ and $B(1, 4, 1)$ $C(0, 5, 1)$ be $Q(\alpha, \beta, \gamma)$. Then $(\alpha^2 + \beta^2 + \gamma^2)$ equal to
[2023 (10 Apr Shift 2)]
(1) 65
(2) 62
(3) 76
(4) 70

18. Let the line $\frac{x}{1} = \frac{6-y}{2} = \frac{z+8}{5}$ intersect the lines $\frac{x-5}{4} = \frac{y-7}{3} = \frac{z+2}{1}$ and $\frac{x+3}{6} = \frac{3-y}{3} = \frac{z-6}{1}$ at the points A and B respectively. Then the distance of the mid-point of the line segment AB from the plane $2x - 2y + z = 14$ is
[2023 (10 Apr Shift 2)]
(1) 3
(2) $\frac{11}{3}$
(3) 4
(4) $\frac{10}{3}$
19. Let the foot of perpendicular from the point $A(4, 3, 1)$ on the plane $P: x - y + 2z + 3 = 0$ be N. If $B(5, \alpha, \beta), \alpha, \beta \in \mathbb{Z}$ is a point on plane P such that the area of the triangle ABN is $3\sqrt{2}$, then $\alpha^2 + \beta^2 + \alpha\beta$ is equal to _____
[2023 (10 Apr Shift 2)]
20. Let (α, β, γ) be the image of point $P(2, 3, 5)$ in the plane $2x + y - 3z = 6$. Then $\alpha + \beta + \gamma$ is equal to
[2023 (11 Apr Shift 1)]
(1) 5
(2) 10
(3) 12
(4) 9
21. If the equation of the plane that contains the point $(-2, 3, 5)$ and is perpendicular to each of the planes $2x + 4y + 5z = 8$ and $3x - 2y + 3z = 5$ is $\alpha x + \beta y + \gamma z + 97 = 0$ then $\alpha + \beta + \gamma =$
[2023 (11 Apr Shift 1)]
(1) 15
(2) 18
(3) 16
(4) 17
22. Let a line L pass through the origin and be perpendicular to the lines $L_1: \vec{r} = (\hat{i} - 11\hat{j} - 7\hat{k}) + \lambda(\hat{i} + 2\hat{j} + 3\hat{k}), \lambda \in \mathbb{R}$ and $L_2: \vec{r} = (-\hat{i} + \hat{k}) + \mu(2\hat{i} + 2\hat{j} + \hat{k}), \mu \in \mathbb{R}$. If P is the point of intersection of L and L_1 , and Q(α, β, γ) is the foot of perpendicular from P on L_2 , then $9(\alpha + \beta + \gamma)$ is equal to _____.
[2023 (11 Apr Shift 1)]
23. Let P be the plane passing through the points $(5, 3, 0), (13, 3, -2)$ and $(1, 6, 2)$. For $\alpha \in \mathbb{N}$, if the distance of the points $A(3, 4, \alpha)$ and $B(2, \alpha, a)$ from the plane P are 2 and 3 respectively, then the positive value of a is
[2023 (11 Apr Shift 2)]
(1) 6
(2) 3
(3) 5
(4) 4
24. Let the line passing through the points $P(2, -1, 2)$ and $Q(5, 3, 4)$ meet the plane $x - y + z = 4$ at the point R. Then the distance of the point R from the plane $x + 2y + 3z + 2 = 0$ measured parallel to the line $\frac{x-7}{2} = \frac{y+3}{2} = \frac{z-2}{1}$ is
[2023 (11 Apr Shift 2)]
(1) $\sqrt{61}$
(2) $\sqrt{189}$
(3) $\sqrt{31}$
(4) 3
25. Let the line $L: x = \frac{1-y}{-2} = \frac{z-3}{\lambda}, \lambda \in \mathbb{R}$ meet the plane $P: x + 2y + 3z = 4$ at the point (α, β, γ) . If the angle between the line L and the plane P is $\cos^{-1}\left(\sqrt{\frac{5}{14}}\right)$, then $\alpha + 2\beta + 6\gamma$ is equal to
[2023 (11 Apr Shift 2)]
26. Let the lines $L_1: \frac{x+5}{3} = \frac{y+4}{1} = \frac{z-\alpha}{-2}$ and $L_2: 3x + 2y + z - 2 = 0 = x - 3y + 2z - 13$ be coplanar. If the point $P(a, b, c)$ on L_1 is nearest to the point $Q(-4, -3, 2)$, then $|a| + |b| + |c|$ is equal to
[2023 (12 Apr Shift 1)]
(1) 12
(2) 14
(3) 8
(4) 10

27. Let the plane $P : 4x - y + z = 10$ be rotated by an angle $\frac{\pi}{2}$ about its line of intersection with the plane $x + y - z = 4$. If α is the distance of the point $(2, 3, -4)$ from the new position of the plane P , then 35α is equal to
[2023 (12 Apr Shift 1)]
(1) 85
(2) 105
(3) 126
(4) 90
28. Let the equation of plane passing through the line of intersection of the planes $x + 2y + az = 2$ and $x - y + z = 3$ be $5x - 11y + bz = 6a - 1$. For $c \in \mathbb{Z}$, if the distance of this plane from the point $(a, -c, c)$ is $\frac{2}{\sqrt{a}}$, then $\frac{a+b}{c}$ is equal to
[2023 (13 Apr Shift 1)]
(1) 2
(2) 4
(3) -4
(4) -2
29. The distance of the point $(-1, 2, 3)$ from the plane $\vec{r} \cdot (\hat{i} - 2\hat{j} + 3\hat{k}) = 10$ parallel to the line of the shortest distance between the lines $\vec{r} = (\hat{i} - \hat{j}) + \lambda(2\hat{i} + \hat{k})$ and $\vec{r} = (2\hat{i} - \hat{j}) + \mu(\hat{i} - \hat{j} + \hat{k})$ is
[2023 (13 Apr Shift 1)]
(1) $4\sqrt{6}$
(2) $2\sqrt{5}$
(3) $2\sqrt{6}$
(4) $3\sqrt{5}$
30. Let the image of the point $(\frac{5}{3}, \frac{5}{3}, \frac{8}{3})$ in the plane $x - 2y + z - 2 = 0$ be P . If the distance of the point $Q(6, -2, \alpha)$, $\alpha > 0$, from P is 13, then α is equal to

[2023 (13 Apr Shift 1)]
31. The plane, passing through the points $(0, -1, 2)$ and $(-1, 2, 1)$ and parallel to the line passing through $(5, 1, -7)$ and $(1, -1, -1)$, also passes through the point
[2023 (13 Apr Shift 2)]
(1) $(-2, 5, 0)$
(2) $(1, -2, 1)$
(3) $(2, 0, 1)$
(4) $(0, 5, -2)$
32. The line, that is coplanar to the line $\frac{x+3}{-3} = \frac{y-1}{1} = \frac{z-5}{5}$, is
[2023 (13 Apr Shift 2)]
(1) $\frac{x+1}{-1} = \frac{y-2}{2} = \frac{z-5}{4}$
(2) $\frac{x+1}{-1} = \frac{y-2}{2} = \frac{z-5}{5}$
(3) $\frac{x-1}{-1} = \frac{y-2}{2} = \frac{z-5}{5}$
(4) $\frac{x+1}{1} = \frac{y-2}{2} = \frac{z-5}{5}$
33. Let N be the foot of perpendicular from the point $P(1, -2, 3)$ on the line passing through the points $(4, 5, 8)$ and $(1, -7, 5)$. Then the distance of N from the plane $2x - 2y + z + 5 = 0$ is
[2023 (13 Apr Shift 2)]
(1) 8
(2) 6
(3) 9
(4) 7
34. Let the foot of perpendicular of the point $P(3, -2, -9)$ on the plane passing through the points $(-1, -2, -3)$, $(9, 3, 4)$, $(9, -2, 1)$ be $Q(\alpha, \beta, \gamma)$. Then the distance Q from the origin is
[2023 (15 Apr Shift 1)]
(1) $\sqrt{42}$
(2) $\sqrt{38}$
(3) $\sqrt{35}$
(4) $\sqrt{29}$

35. Let S be the set of all values of λ , for which the shortest distance between the lines $\frac{x-\lambda}{0} = \frac{y-3}{4} = \frac{z+6}{1}$ and $\frac{x+\lambda}{3} = \frac{y}{-4} = \frac{z-6}{0}$ is 13. Then $8|\sum_{\lambda \in S} \lambda|$ is equal to
[2023 (15 Apr Shift 1)]
(1) 306
(2) 304
(3) 308
(4) 302
36. If the line $x = y = z$ intersects the line $x \sin A + y \sin B + z \sin C - 18 = 0 = x \sin 2A + y \sin 2B + z \sin 2C - 9$, where A, B, C are the angles of a triangle ABC , then $80 \left(\sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} \right)$ is equal to _____.
[2023 (15 Apr Shift 1)]
37. Let the plane P contain the line $2x + y - z - 3 = 0 = 5x - 3y + 4z + 9$ and be parallel to the line $\frac{x+2}{2} = \frac{3-y}{-4} = \frac{z-7}{5}$. Then the distance of the point $A(8, -1, -19)$ from the plane P measured parallel to the line $\frac{x}{-3} = \frac{y-5}{4} = \frac{2-z}{-12}$ is equal to _____.
[2023 (15 Apr Shift 1)]

ANSWER KEYS

- | | | | | | | | |
|----------|---------|----------|----------|----------|----------|---------|---------|
| 1. (4) | 2. (2) | 3. (594) | 4. (1) | 5. (1) | 6. (18) | 7. (4) | 8. (2) |
| 9. (9) | 10. (1) | 11. (1) | 12. (11) | 13. (1) | 14. (3) | 15. (4) | 16. (2) |
| 17. (1) | 18. (3) | 19. (7) | 20. (2) | 21. (1) | 22. (5) | 23. (4) | 24. (4) |
| 25. (11) | 26. (4) | 27. (3) | 28. (3) | 29. (3) | 30. (15) | 31. (1) | 32. (2) |
| 33. (4) | 34. (1) | 35. (1) | 36. (5) | 37. (26) | | | |

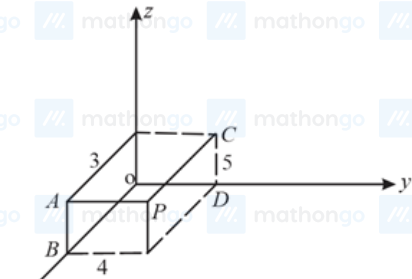
1. (4)

Given,

One vertex of a rectangular parallelepiped is at the origin O and the lengths of its edges along x , y and z axes are 3, 4 and 5 units respectively,

And P be the vertex (3, 4, 5),

Now plotting the diagram we get,



Now equation of line OP : $\frac{x}{3} = \frac{y}{4} = \frac{z}{5}$

And equation of line AB : $\frac{x-3}{0} = \frac{y}{0} = \frac{z}{1}$

Now finding, $\vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 4 & 5 \\ 0 & 0 & 1 \end{vmatrix}$

$\Rightarrow \vec{n}_1 \times \vec{n}_2 = \hat{i}(4) - \hat{j}(3) + \hat{k}(0)$

$\Rightarrow \vec{n}_1 \times \vec{n}_2 = 4\hat{i} - 3\hat{j}$

Now we know that,

Shortest Distance, S. D = $\frac{(\vec{a}_2 - \vec{a}_1) \cdot (\vec{n}_1 \times \vec{n}_2)}{|\vec{n}_1 \times \vec{n}_2|}$

$\Rightarrow \text{S. D} = \frac{(3\hat{i} + 0\hat{j} + 0\hat{k} - 0\hat{i} + 0\hat{j} + 0\hat{k}) \cdot (4\hat{i} - 3\hat{j})}{5}$

$\Rightarrow \text{S. D} = \frac{(3\hat{i}) \cdot (4\hat{i} - 3\hat{j})}{5}$

$\Rightarrow \text{S. D} = \frac{12}{5}$

2. (2)

Given,

The equation of the plane passing through the line of intersection of the planes $2x - y + z = 3$, $4x - 3y + 5z + 9 = 0$ and parallel to the line $\frac{x+1}{-2} = \frac{y+3}{4} = \frac{z-2}{5}$ is $ax + by + cz + 6 = 0$,

Now let, $P_1 = 2x - y + z = 3$

And $P_2 = 4x - 3y + 5z + 9 = 0$

Now using family of planes we get,

$$P_1 + \lambda P_2 = 0$$

$$\Rightarrow (2x - y + z - 3) + \lambda(4x - 3y + 5z + 9) = 0$$

$$\Rightarrow (2 + 4\lambda)x - (1 + 3\lambda)y + (1 + 5\lambda)z - (3 - 9\lambda) = 0$$

Now let,

$$P_3 = (2 + 4\lambda)x - (1 + 3\lambda)y + (1 + 5\lambda)z - (3 - 9\lambda) = 0$$

Now given P_3 is parallel to line,

$$\frac{x+1}{-2} = \frac{y+3}{4} = \frac{z-2}{5}$$

So, using perpendicular condition as normal vector of plane will be perpendicular to given line we get,

$$-2(2 + 4\lambda) - 4(1 + 3\lambda) + 5(1 + 5\lambda) = 0$$

$$\Rightarrow -3 + 5\lambda = 0$$

$$\Rightarrow \lambda = \frac{3}{5}$$

Hence, equation of plane P_3 will be,

$$P_3 : \frac{22x}{5} - \frac{14y}{5} + \frac{20z}{5} + \frac{12}{5} = 0$$

$$\Rightarrow P_3 = 11x - 7y + 10z + 6 = 0$$

Now on comparing with $ax + by + cz + 6 = 0$, we get,

$$a = 11, b = -7 \text{ and } c = 10$$

$$\therefore a + b + c = 11 - 7 + 10 = 14$$

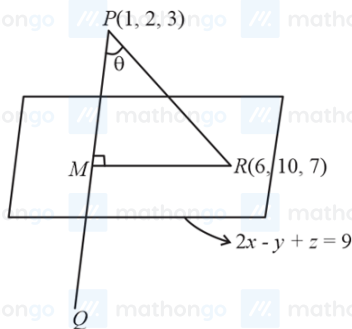
3. (594)

Given,

The image of the point $P(1, 2, 3)$ in the plane $2x - y + z = 9$ be Q ,

The coordinates of the point R are $(6, 10, 7)$,

Now plotting the diagram we get,



R lies on plane as $(6, 10, 7)$ satisfy the plane equation $2x - y + z = 9$

Now length of $PR = \sqrt{5^2 + 8^2 + 4^2} = \sqrt{105}$

Now finding the angle between PR & PM (which is normal vector to plane) we get,

$$\cos \theta = \frac{\vec{PR} \cdot \vec{PM}}{|\vec{PR}| \cdot |\vec{PM}|}$$

$$\Rightarrow \cos \theta = \frac{(5\hat{i} + 8\hat{j} + 4\hat{k}) \cdot (2\hat{i} - \hat{j} + \hat{k})}{\sqrt{105} \sqrt{6}}$$

$$\Rightarrow \cos \theta = \frac{6}{\sqrt{630}}$$

Now $\text{Area}(\Delta PQR) = 2 \text{ area}(\Delta PMR)$

$$\Rightarrow \Delta PQR = 2 \cdot \frac{1}{2} \cdot PR \sin \theta \cdot PR \cos \theta$$

$$\Rightarrow \Delta PQR = 2 \cdot \frac{1}{2} (PR)^2 \sin \theta \cos \theta$$

$$\Rightarrow \Delta PQR = 105 \cdot \frac{6}{\sqrt{630}} \cdot \frac{\sqrt{594}}{\sqrt{630}}$$

$$\Rightarrow \Delta PQR = \sqrt{594}$$

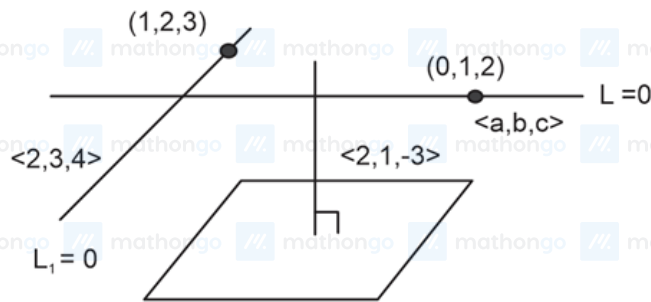
Hence, area of the square will be 594

4. (1)

Given,

The line L pass through the point $(0, 1, 2)$, intersect the line $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and be parallel to the plane $2x + y - 3z = 4$,

Now plotting the diagram of above data we have,



Now let the direction of line which passes through $(0, 1, 2)$ be (a, b, c)

Now the line L & L_1 are coplanar,

So coplanar condition we get,

$$\begin{vmatrix} a & b & c \\ 2 & 1 & 3 \\ 1 & 2 & 4 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} a & b & c \\ 1 & 1 & 1 \\ 2 & 3 & 4 \end{vmatrix} = 0$$

$$\Rightarrow a - 2b + c = 0 \dots\dots(1)$$

And the line L is perpendicular to the normal vector of the plane,

So by perpendicular condition we get,

$$2a + b - 3c = 0 \dots\dots(2)$$

So, from equation (1) & (2) we get,

$$a = b = c$$

So, equation of line L will be,

$$\Rightarrow L = \frac{x}{1} = \frac{y-1}{1} = \frac{z-2}{1} = \lambda$$

So any point on L can be taken as $A(\lambda, 1 + \lambda, 2 + \lambda)$,

$$A(\lambda, 1 + \lambda, 2 + \lambda)$$



$$P(1, -9, 2)$$

Now $\vec{AP}$ will be perpendicular to the direction ratio of the line L ,

So, again by perpendicular condition we get,

$$\vec{AP} \cdot (\hat{i} + \hat{j} + \hat{k}) = 0$$

$$\Rightarrow \lambda - 1 + \lambda + 10 + \lambda = 0$$

$$\Rightarrow 3\lambda + 9 = 0$$

$$\Rightarrow \lambda = -3$$

Hence, $A(-3, -2, -1)$ & $P(1, -9, 2)$

So, by distance formula we get, $AP = \sqrt{74}$

5. (1)

Given,

A plane P contains the line of intersection of the plane $\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = 6$ and $\vec{r} \cdot (2\hat{i} + 3\hat{j} + 4\hat{k}) = -5$,

Now let, $P_1 : x + y + z - 6 = 0$

And $P_2 : 2x + 3y + 4z + 5 = 0$

So, by family of planes concept we get,

$P : (x + y + z - 6) + \lambda(2x + 3y + 4z + 5) = 0$

Also given plane P passes through $(0, 2, -2)$

So, $(0 + 2 - 2 - 6) + \lambda(4 + 6 - 8 + 5) = 0$

$\Rightarrow -6 + \lambda(3) = 0$

$\Rightarrow \lambda = 2$

Hence, equation of plane will be,

$P : 5x + 7y + 9z + 4 = 0$

Now distance of point $(12, 12, 18)$ from plane will be, $\left(\frac{5(12) + 7(12) + 9(18) + 4}{\sqrt{25 + 49 + 81}} \right)$

And square of distance will be,

$$= \left(\frac{5(12) + 7(12) + 9(18) + 4}{\sqrt{25 + 49 + 81}} \right)^2$$

$$= \frac{310 \times 310}{155} = 620$$

6. (18)

Let,

$$L_1 = \frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{\alpha} = \lambda$$

$$L_2 = \frac{x-4}{5} = \frac{y-1}{2} = \frac{z-0}{\beta} = \mu$$

For point of intersection

$$2\lambda + 1 = 5\mu + 4 \quad \dots (i)$$

$$3\lambda + 2 = 2\mu + 1 \quad \dots (ii)$$

$$\alpha\lambda + 3 = \beta\mu + 0 \quad \dots (iii)$$

From (i) and (ii), we get $\lambda = \mu = -1$

Now, from (iii), we get $\alpha - \beta = 3$

Let $y = 8\alpha\beta$

$$\Rightarrow y = 8\alpha(\alpha - 3)$$

$$\Rightarrow y = 8\left(\alpha^2 - 3\alpha + \frac{9}{4}\right) - 18$$

$$\Rightarrow y = 8\left(\alpha - \frac{3}{2}\right)^2 - 18$$

So, Minimum value of $y = -18$ at $\alpha = +\frac{3}{2}$

7. (4)

Given:

$$\frac{x-4}{4} = \frac{y+2}{5} = \frac{z+3}{3}$$

$$\frac{x-1}{3} = \frac{y-3}{4} = \frac{z-4}{2}$$

So,

$$\vec{a}_1 = 4\hat{i} - 2\hat{j} - 3\hat{k}$$

$$\vec{a}_2 = \hat{i} + 3\hat{j} + 4\hat{k}$$

So,

$$\vec{a}_2 - \vec{a}_1 = -3\hat{i} + 5\hat{j} + 7\hat{k}$$

And,

$$\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & 5 & 3 \\ 3 & 4 & 2 \end{vmatrix} = -2\hat{i} + \hat{j} + \hat{k}$$

$$\Rightarrow |\vec{b}_1 \times \vec{b}_2| = \sqrt{(-2)^2 + (1)^2 + (1)^2} = \sqrt{6}$$

Shortest distance between the lines

$$= \frac{|(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|}$$

$$= \frac{|(-3\hat{i} + 5\hat{j} + 7\hat{k}) \cdot (-2\hat{i} + \hat{j} + \hat{k})|}{\sqrt{6}}$$

$$= \frac{|6 + 5 + 7|}{\sqrt{6}} = \frac{6 \times 3}{\sqrt{6}} \text{ units}$$

$$= 3\sqrt{6} \text{ units}$$

Hence this is the correct option.

8. (2)

Given,

The equation of the plane containing the line $x + 2y + 3z - 4 = 0 = 2x + y - z + 5$ and perpendicular to the plane

$$\vec{r} = (\hat{i} - \hat{j}) + \lambda(\hat{i} + \hat{j} + \hat{k}) + \mu(\hat{i} - 2\hat{j} + 3\hat{k}) \text{ is } ax + by + cz = 4,$$

Now let equation of required plane be,

$$P : (x + 2y + 3z - 4) + \alpha(2x + y - z + 5) = 0$$

$$\Rightarrow (2\alpha + 1)x + (\alpha + 2)y + (3 - \alpha)z = 4 - 5\alpha$$

So, the normal vector is given by,

$$\vec{n}_1 = (2\alpha + 1)\hat{i} + (\alpha + 2)\hat{j} + (3 - \alpha)\hat{k}$$

And normal of the plane $\vec{r} = (\hat{i} - \hat{j}) + \lambda(\hat{i} + \hat{j} + \hat{k}) + \mu(\hat{i} - 2\hat{j} + 3\hat{k})$ will be,

$$\vec{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 1 & -2 & 3 \end{vmatrix} = 5\hat{j} - 2\hat{j} - 3\hat{k}$$

Now given they are perpendicular, so by perpendicular condition we get,

$$\vec{n}_1 \cdot \vec{n}_2 = 0$$

$$\Rightarrow 5(2\alpha + 1) - 2(\alpha + 2) - 3(3 - \alpha) = 0$$

$$\Rightarrow 11\alpha + (-8) = 0$$

$$\Rightarrow \alpha = \frac{8}{11}$$

Hence, equation of required plane will be,

$$P : \frac{27}{11}x + \frac{30}{11}y + \frac{25}{11}z - \frac{4}{11} = 0$$

$$\Rightarrow 27x + 30y + 25z = 4$$

Now comparing with $ax + by + cz = 4$ we get,

$$a = 27, b = 30, c = 25$$

$$\text{Hence, } a - b + c = 22$$

9. (9)

Given,

λ_1, λ_2 be the values of λ for which the points $(\frac{5}{2}, 1, \lambda)$ and $(-2, 0, 1)$ are at equal distance from the plane $2x + 3y - 6z + 7$,

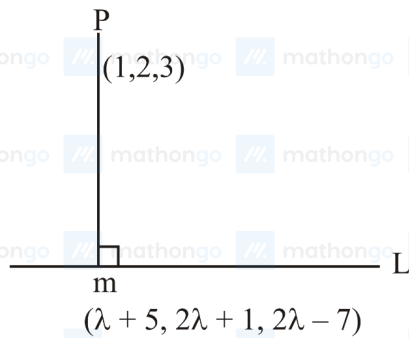
So, by using distance formula of a point from a plane we get,

$$\begin{aligned} \left| \frac{-4+0-6+7}{\sqrt{2^2+3^2+6^2}} \right| &= \left| \frac{5+3-6\lambda+7}{\sqrt{2^2+3^2+6^2}} \right| \\ \Rightarrow \left| \frac{-4+0-6+7}{7} \right| &= \left| \frac{5+3-6\lambda+7}{7} \right| \\ \Rightarrow \frac{3}{7} &= \left| \frac{15-6\lambda}{7} \right| \\ \Rightarrow \lambda &= 2 \text{ or } 3 \end{aligned}$$

$$\Rightarrow \lambda_1 = 3, \lambda_2 = 2 \text{ as } \lambda_1 > \lambda_2$$

$$\text{So, } (\lambda_1 - \lambda_2, \lambda_2, \lambda_1) = (1, 2, 3)$$

Now finding the distance of point $(1, 2, 3)$ from the line $\frac{x-5}{1} = \frac{y-1}{2} = \frac{z+7}{2} = \lambda$ we get,



Now from diagram we can see that, $\overrightarrow{PM}$ is perpendicular to given line,

So, by perpendicular condition we get,

$$\overrightarrow{PM} \cdot (\hat{i} + 2\hat{j} + 2\hat{k}) = 0$$

$$\Rightarrow (\lambda + 4) + 2(2\lambda - 1) + 2(2\lambda - 10) = 0$$

$$\Rightarrow 9\lambda = 18 \text{ or } \lambda = 2$$

Hence, the point M will be $M(7, 5, -3)$

$$\text{So, the distance } |\overrightarrow{PM}| = \sqrt{6^2 + 3^2 + 6^2} = 9$$

10. (1)

Given,

For $a, b \in \mathbb{Z}$ and $|a - b| \leq 10$, let the angle between the plane $P: ax + y + z = b$ and the line $L: x - 1 = a - y = z + 1$ be $\cos^{-1}\left(\frac{1}{3}\right)$

Now rewriting the equation of line we get,

$$L: \frac{x-1}{1} = \frac{y-a}{-1} = \frac{z+1}{1}$$

$$\text{Now let, } \theta = \cos^{-1}\left(\frac{1}{3}\right) = \sin^{-1}\left(\frac{2\sqrt{2}}{3}\right)$$

Now we know that,

Angle between plane and line is given by,

$$\cos(90^\circ - \theta) = \frac{a+1(-1)+(-1)1}{\sqrt{a^2+2}\sqrt{3}}$$

$$\Rightarrow \frac{a-2}{\sqrt{a^2+2}} = \frac{2\sqrt{2}}{\sqrt{3}}$$

$$\Rightarrow 3(a-2)^2 = 8(a^2+2)$$

$$\Rightarrow 3(a^2+4-4a) = 8(a^2+2)$$

$$\Rightarrow 5a^2+12a+4=0$$

$$\text{So, } a = -\frac{2}{5}, a = -2$$

Hence, $a = -2$ ($\because a \in \mathbb{Z}$)

So, the equation of plane will be,

$$P: -2x + y - z - b = 0$$

Now given, distance of point $(6, -6, 4)$ from plane is $= 3\sqrt{6}$

$$\Rightarrow \left| \frac{-12-6-4-b}{\sqrt{2^2+1^2+1^2}} \right| = 3\sqrt{6}$$

$$\Rightarrow |22+b| = 18$$

$$\Rightarrow b = -4, -40$$

So, $b = -4$ ($\because |a-b| \leq 10$)

$$\text{Hence, } a^2 + b^2 = 16 + 16 = 32$$

11. (1)

Given,

P be the plane passing through the line $\frac{x-1}{1} = \frac{y-2}{-3} = \frac{z+5}{7}$ and the point $(2, 4, -3)$,

So, equation of plane is given by,

$$\begin{vmatrix} x-2 & y-4 & z+3 \\ 1 & -3 & 7 \\ 2-1 & 4-2 & -3-(-5) \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} x-2 & y-4 & z+3 \\ 1 & -3 & 7 \\ 1 & 2 & 2 \end{vmatrix} = 0$$

$$\Rightarrow (x-2)(-20) - (y-4)(-5) + (z+3)(5) = 0$$

$$\Rightarrow 4x - y - z - 7 = 0$$

Now finding the image (α, β, γ) of the point $(-1, 3, 4)$ with respect to plane $4x - y - z - 7 = 0$ we get,

$$\frac{x+1}{4} = \frac{y-3}{-1} = \frac{z-4}{-1} = -2 \left(\frac{-4-3-4-7}{16+1+1} \right) = 2$$

$$\Rightarrow x = 7 = \alpha, y = 1 = \beta, z = 2 = \gamma$$

$$\text{Hence, } \alpha + \beta + \gamma = 10$$

12. (11)

Given,

P_1 be the plane $3x - y - 7z = 11$,

P_2 be the plane passing through the points $(2, -1, 0)$, $(2, 0, -1)$, and $(5, 1, 1)$,

Now finding the equation of plane P_2 , we get

$$\begin{vmatrix} x-2 & y+1 & z \\ 2-2 & 0-(-1) & -1-0 \\ 5-2 & 1-0 & 1-(-1) \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} x-2 & y+1 & z \\ 0 & 1 & -1 \\ 3 & 1 & 2 \end{vmatrix} = 0$$

$$\Rightarrow 3(x-2) + 3(-y-1) - 3z = 0$$

$$\Rightarrow x - 2 - y - 1 - z = 0$$

$$\Rightarrow x - y - z = 3$$

Now direction ratio of line of intersection of planes $3x - y - 7z = 11$ & $x - y - z = 3$ is given by,

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -1 & -7 \\ 1 & -1 & -1 \end{vmatrix} \\ = \hat{i}(1-7) - \hat{j}(-3+7) + \hat{k}(-3+1) \\ = -6\hat{i} - 4\hat{j} - 2\hat{k}$$

Or direction ratio is given by $3\hat{i} + 2\hat{j} + \hat{k}$

Now finding the point on the line of intersection by taking $z = 0$ in both plane,

We get, $3x - y = 11$ & $x - y = 3$

Now solving above two lines we get,

$(4, 1, 0)$ is a point on line of intersection

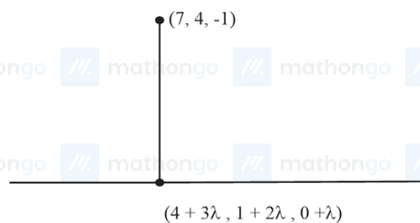
Hence, the equation of line will be,

$$\frac{x-4}{3} = \frac{y-1}{2} = \frac{z}{1} = \lambda$$

Now any point on the line is given by,

$$(3\lambda + 4, 2\lambda + 1, \lambda)$$

Now finding the distance of the point $(7, 4, -1)$ from the line,



Now from above diagram using perpendicular condition we get,

$$3(3\lambda - 3) + 2(2\lambda - 3) + 1(\lambda + 1) = 0$$

$$\Rightarrow 14\lambda = 14 \Rightarrow \lambda = 1$$

Hence, the point is $(7, 3, 1)$ and comparing with (α, β, γ) we get, $\alpha + \beta + \gamma = 11$

13. (1)

Given,

O be the origin and the position vector of the point P be $-\hat{i} - 2\hat{j} + 3\hat{k}$,

So, $\vec{OP} = -\hat{i} - 2\hat{j} + 3\hat{k}$

Also given the position vectors of the points A , B and C are $-2\hat{i} + \hat{j} - 3\hat{k}$, $2\hat{i} + 4\hat{j} - 2\hat{k}$ and $-4\hat{i} + 2\hat{j} - \hat{k}$

Now finding,

$$\vec{AB} = 4\hat{i} + 3\hat{j} + \hat{k}$$

$$\vec{AC} = -2\hat{i} + \hat{j} + 2\hat{k}$$

$$\text{And } \vec{AB} \times \vec{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & 3 & 1 \\ -2 & 1 & 2 \end{vmatrix}$$

$$\Rightarrow \vec{AB} \times \vec{AC} = 5\hat{i} - 10\hat{j} + 10\hat{k}$$

Now finding the Projection of $\vec{OP}$ on vector perpendicular to $\vec{AB}$ & $\vec{AC}$ we get,

$$\text{Projection} = \frac{|\vec{OP} \cdot (\vec{AB} \times \vec{AC})|}{|\vec{AB} \times \vec{AC}|}$$

$$= \frac{5(-\hat{i} - 2\hat{j} + 3\hat{k}) \cdot (5\hat{i} - 10\hat{j} + 10\hat{k})}{5\sqrt{1+4+4}} = 3$$

14. (3)

Given,

Two vertices of a triangle ABC be $(2, 4, 6)$ and $(0, -2, -5)$, and its centroid be $(2, 1, -1)$

Now, let the vertex ' C ' be (a, b, c) and using centroid formula we get,

$$2 = \frac{2+0+a}{3} \Rightarrow a = 4$$

$$1 = \frac{4-2+b}{3} \Rightarrow b = 1$$

$$-1 = \frac{6-5+c}{3} \Rightarrow c = -4$$

So, the third vertex will be, $C(4, 1, -4)$

Now finding image of C in $x + 2y + 4z = 11$ we get,

$$\frac{\alpha-4}{1} = \frac{\beta-1}{2} = \frac{\gamma+4}{4} = -2 \frac{(4+2-16-11)}{16+4+1}$$

$$\Rightarrow \frac{\alpha-4}{1} = 2 \Rightarrow \alpha = 6, \frac{\beta-1}{2} = 2 \Rightarrow \beta = 5 \text{ and } \frac{\gamma+4}{4} = 2 \Rightarrow \gamma = 4$$

Now find the required answer we get,

$$\alpha\beta + \beta\gamma + \alpha\gamma = 30 + 20 + 24 = 74$$

15. (4)

Given,

$$\frac{x+2}{1} = \frac{y}{-2} = \frac{z-5}{2}$$

$$\frac{x-4}{1} = \frac{y-1}{2} = \frac{z+3}{0}$$

Now we know that,

Shortest distance between two skew lines is given by,

$$SD = \frac{|(\vec{a}_2 - \vec{a}_1) \cdot (\vec{n}_1 \times \vec{n}_2)|}{|\vec{n}_1 \times \vec{n}_2|}$$

$$\text{Now finding, } \vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 2 \\ 1 & 2 & 0 \end{vmatrix}$$

$$\Rightarrow \vec{n}_1 \times \vec{n}_2 = -4\hat{i} + 2\hat{j} + 4\hat{k}$$

$$\text{And } \vec{a}_2 - \vec{a}_1 = (4\hat{i} + \hat{j} - 3\hat{k}) - (-2\hat{i} + 0\hat{j} + 5\hat{k}) = 6\hat{i} + \hat{j} - 8\hat{k}$$

Now putting the value in formula we get,

$$SD = \frac{|(6\hat{i} + \hat{j} - 8\hat{k}) \cdot (-4\hat{i} + 2\hat{j} + 4\hat{k})|}{\sqrt{16+16+4}}$$

$$\Rightarrow SD = \left| \frac{-24+2-32}{6} \right| = \frac{54}{6} = 9$$

16. (2)

Given,

P be the point of intersection of the line $\frac{x+3}{3} = \frac{y+2}{1} = \frac{1-z}{2}$ and the plane $x + y + z = 2$

Now let, $\frac{x+3}{3} = \frac{y+2}{1} = \frac{1-z}{2} = \lambda$

Say, point $A(3\lambda - 3, \lambda - 2, 1 - 2\lambda)$

Since point lie on the plane also,

So, $3\lambda - 3 + \lambda - 2 + 1 - 2\lambda = 2$

$$\Rightarrow 2\lambda = 6$$

$$\Rightarrow \lambda = 3$$

Hence, the point will be $P(6, 1, -5)$

Now finding the distance of the point $P(6, 1, -5)$ from the plane $3x - 4y + 12z = 32$ we get,

$$q = \left| \frac{18 - 4 - 60 - 32}{\sqrt{9 + 16 + 144}} \right| = \frac{78}{13} = 6$$

Equation with roots q and $2q$ will be,

$$x^2 - 3qx + 2q^2 = 0$$

$$\Rightarrow x^2 - 18x + 72 = 0$$

17. (1)

Plane passes through the points

$A(1, 2, 0)$, $B(1, 4, 1)$ and $C(0, 5, 1)$

We know that the equation of a plane passing through three non-collinear points (x_1, y_1, z_1) , (x_2, y_2, z_2) , (x_3, y_3, z_3) is

$$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ x_3 - x_1 & y_3 - y_1 & z_3 - z_1 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} x - 1 & y - 2 & z - 0 \\ 1 - 1 & 4 - 2 & 1 - 0 \\ 0 - 1 & 5 - 2 & 1 - 0 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} x - 1 & y - 2 & z - 0 \\ 0 & 2 & 1 \\ -1 & 3 & 1 \end{vmatrix} = (x - 1)(2 - 3) - (y - 2)(0 + 1) + z(0 + 2) = 0$$

$$\Rightarrow -x + 1 - y + 2 + 2z = 0$$

$$\Rightarrow x + y - 2z = 3 \quad \dots\dots (i)$$

Now image of $P(1, 2, 6)$ in (i)

$$\frac{x-1}{1} = \frac{y-2}{1} = \frac{z-6}{-2} = \frac{-2(-9-3)}{6} = 4$$

$$x = 5, y = 6, z = -2$$

$$\Rightarrow \alpha = 5, \beta = 6, \gamma = -2$$

$$\therefore \alpha^2 + \beta^2 + \gamma^2 = 65$$

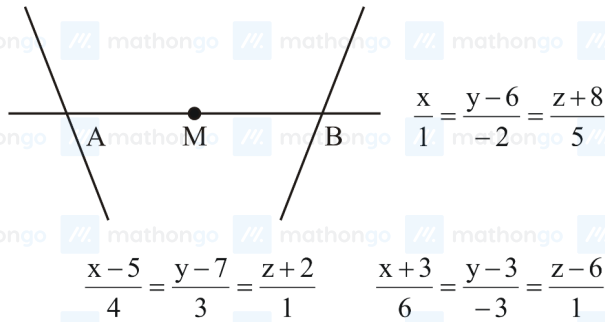
Hence this is the required option.

18. (3)

Given,

The line $\frac{x}{1} = \frac{y-6}{-2} = \frac{z+8}{5}$ intersect the lines $\frac{x-5}{4} = \frac{y-7}{3} = \frac{z+2}{1}$ and $\frac{x+3}{6} = \frac{y-3}{-3} = \frac{z-6}{1}$ at the points A and B respectively,

Now plotting the diagram, we get,



Now solving, $\frac{x}{1} = \frac{y-6}{-2} = \frac{z+8}{5} = \lambda$ and $\frac{x-5}{4} = \frac{y-7}{3} = \frac{z+2}{1} = t$ we get,

$$A \equiv (\lambda, -2\lambda + 6, 5\lambda - 8) \equiv (4t + 5, 3t + 7, t - 2)$$

Now comparing both side and solving we get, $t = -1, \lambda = 1$

Hence, point $A(1, 4, -3)$

Now solving, $\frac{x}{1} = \frac{y-6}{-2} = \frac{z+8}{5} = \mu$ and $\frac{x+3}{6} = \frac{y-3}{-3} = \frac{z-6}{1} = k$ we get,

$$\text{For point } B \equiv (\mu, -2\mu + 6, 5\mu - 8) \equiv (6k - 3, -3k + 3, k + 6)$$

Now comparing both side and solving we get, $k = 1, \mu = 3$

Hence, point $B(3, 0, 7)$

So, mid-point of $AB = M(2, 2, 2)$

Now finding, distance of M from the plane $2x - 2y + z - 14 = 0$ we get,

$$\left| \frac{4 - 4 + 2 - 14}{\sqrt{2^2 + 2^2 + 1^2}} \right| = \left| \frac{-12}{3} \right| = 4$$

19. (7)

Given,

The foot of perpendicular from the point $A(4, 3, 1)$ on the plane $P: x - y + 2z + 3 = 0$ be N

So, by foot of perpendicular formula we get,

$$\frac{x_1 - 4}{1} = \frac{y_1 - 3}{-1} = \frac{z_1 - 1}{2} = -\frac{4 - 3 + 2 + 3}{1^2 + 1^2 + 2^2}$$

$$\Rightarrow N \equiv (x_1, y_1, z_1) = (3, 4, -1)$$

And $B(5, \alpha, \beta), \alpha, \beta \in \mathbb{Z}$ is a point on plane P such that the area of the triangle ABN is $3\sqrt{2}$,

So putting the value of point $B(5, \alpha, \beta)$ in plane we get,

$$5 - \alpha + 2\beta + 3 = 0$$

$$\Rightarrow \alpha - 2\beta = 8$$

And given area of triangle ABN is $3\sqrt{2}$, we get

$$\frac{1}{2} AN \times BN = 3\sqrt{2}$$

$$\Rightarrow AN \times BN = 6\sqrt{2}$$

$$\Rightarrow \frac{6}{\sqrt{6}} \times BN = 6\sqrt{2}$$

$$\Rightarrow BN = 2\sqrt{3}$$

$$\Rightarrow 4 + (\alpha - 4)^2 + (\beta + 1)^2 = 12$$

$$\Rightarrow (\alpha - 4)^2 + (\beta + 1)^2 = 8$$

$$\Rightarrow 5\beta^2 + 18\beta + 9 = 0 \text{ \{as } \alpha - 2\beta = 8\}}$$

$$\Rightarrow \beta = -3 \text{ \{as } \beta \neq -\frac{3}{5}\}}$$

So, $\beta = -3, \alpha = 2$

$$\text{Hence, } \alpha^2 + \beta^2 + \alpha\beta = 3^2 + 2^2 - 6 = 7$$

20. (2)

Given plane is $2x + y - 3z = 6$.

Since, (α, β, γ) is the image point of $(2, 3, 5)$ in the given plane, so

$$\frac{\alpha-2}{2} = \frac{\beta-3}{1} = \frac{\gamma-5}{-3} = -\frac{2(4+3-15-6)}{4+1+9}$$

$$\Rightarrow \frac{\alpha-2}{2} = \frac{\beta-3}{1} = \frac{\gamma-5}{-3} = 2$$

$$\Rightarrow \frac{\alpha-2}{2} = 2; \frac{\beta-3}{1} = 2; \frac{\gamma-5}{-3} = 2$$

$$\therefore \alpha = 6, \beta = 5, \gamma = -1$$

$$\text{Hence, } \alpha + \beta + \gamma = 6 + 5 - 1 = 10$$

Hence this is the required option.

21. (1)

Given

The equation of the plane that contains the point $(-2, 3, 5)$ and is perpendicular to each of the planes $2x + 4y + 5z = 8$ and $3x - 2y + 3z = 5$ is

$$\alpha x + \beta y + \gamma z + 97 = 0$$

Now satisfying the point $(-2, 3, 5)$ in plane $P : \alpha x + \beta y + \gamma z + 97 = 0$ we get,

$$-2\alpha + 3\beta + 5\gamma + 97 = 0 \dots\dots(i)$$

And using perpendicular condition $2x + 4y + 5z = 8$ and $\alpha x + \beta y + \gamma z + 97 = 0$ we get,

$$2\alpha + 4\beta + 5\gamma = 0 \dots\dots(ii)$$

$$\text{Similarly, } 3\alpha - 2\beta + 3\gamma = 0 \dots\dots(iii)$$

Now solving equation (i), (ii) & (iii) we get,

$$\alpha = 22, \beta = 9 \text{ \& } \gamma = -16$$

$$\text{Hence, } \alpha + \beta + \gamma = 22 + 9 - 16 = 15$$

22. (5)

Given,

A line L pass through the origin and be perpendicular to the lines $L_1 : \vec{r} = (\hat{i} - 11\hat{j} - 7\hat{k}) + \lambda(\hat{i} + 2\hat{j} + 3\hat{k}), \lambda \in \mathbb{R}$ and $L_2 : \vec{r} = (-\hat{i} + \hat{k}) + \mu(2\hat{i} + 2\hat{j} + \hat{k}), \mu \in \mathbb{R}$

So, direction ratio of line will be,

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 3 \\ 2 & 2 & 1 \end{vmatrix}$$

$$= \hat{i}(-4) - \hat{j}(-5) + \hat{k}(-2) \\ = -4\hat{i} + 5\hat{j} - 2\hat{k}$$

Hence, the equation of line L which passes through origin will be,

$$L : \vec{r} = \sigma(-4\hat{i} + 5\hat{j} - 2\hat{k})$$

Now finding the intersection point P we get,

$$1 + \lambda = -4\sigma \dots (1)$$

$$-11 + 2\lambda = 5\sigma \dots (2)$$

$$-7 + 3\lambda = -2\sigma \dots (3)$$

So, from equation (1) & (3) we get, $1 + \lambda = -14 + 6\lambda \Rightarrow \lambda = 3, \sigma = -1$

Hence, point $P \equiv (4, -5, 2)$

Now finding the point Q which is foot of perpendicular from point $P(4, -5, 2)$ on line $L_2 : (-\hat{i} + \hat{k}) + \mu(2\hat{i} + 2\hat{j} + \hat{k})$ we get,

Now any point on line L_2 will be $Q((2\mu - 1)\hat{i} + 2\mu\hat{j} + (\mu + 1)\hat{k})$

Now direction ratio of PQ will be $(-5 + 2\mu)\hat{i} + (2\mu + 5)\hat{j} + (\mu - 1)\hat{k} = 0$

Now using the perpendicular condition we get,

$$2(-5 + 2\mu) + 2(2\mu + 5) + 1(\mu - 1) = 0$$

$$\Rightarrow \mu = \frac{1}{9}$$

$P(4, -5, 2)$

$M(-1 + 2\mu, 2\mu, 1 + \mu)$

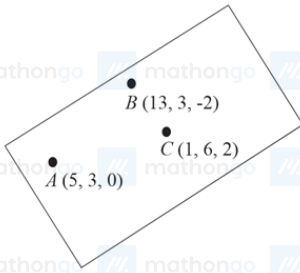
$$\text{Hence, } \alpha + \beta + \gamma = 2\mu - 1 + 2\mu + \mu + 1 = 5\mu = \frac{5}{9}$$

$$\Rightarrow 9(\alpha + \beta + \gamma) = 5$$

23. (4)

Given,

P be the plane passing through the points $(5, 3, 0)$, $(13, 3, -2)$ and $(1, 6, 2)$



Now, $\vec{AB} = 8\hat{i} - 2\hat{k}$, $\vec{AC} = -4\hat{i} + 3\hat{j} + 2\hat{k}$,

So, normal vector to the plane will be,

$$\vec{AB} \times \vec{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 8 & 0 & -2 \\ -4 & 3 & 2 \end{vmatrix} = 6\hat{i} - 8\hat{j} + 24\hat{k}$$

Now equation of plane will be: $6x - 8y + 24z = d$

And given plane passes through $(5, 3, 0)$

So, $6 \times 5 - 8 \times 3 + 24 \times 0 = d \Rightarrow d = 6$

Hence, equation of plane will be,

$$6x - 8y + 24z = 6 \Rightarrow 3x - 4y + 12z = 3$$

Now using the formula distance of point $(3, 4, \alpha)$ from the plane $3x - 4y + 12z = 3$, we get,

$$\frac{9 - 16 + 12\alpha - 3}{\sqrt{9 + 16 + 144}} = 2 \Rightarrow \alpha = 3$$

Now again using the distance of point $(2, \alpha, a) \equiv (2, 3, a)$ from the plane formula we get,

$$\frac{3 \times 2 - 4 \times 3 + 12 \times a - 3}{13} = 3$$

$$\Rightarrow 12a - 9 = 39$$

$$\Rightarrow 12a = 48$$

$$\Rightarrow a = 4$$

24. (4)

Given,

The line passing through the points $P(2, -1, 2)$ and $Q(5, 3, 4)$,

So, equation of line PQ will be,

$$\frac{x-2}{3} = \frac{y+1}{4} = \frac{z-2}{2} = \lambda$$

Now let point R be $(3\lambda + 2, 4\lambda - 1, 2\lambda + 2)$

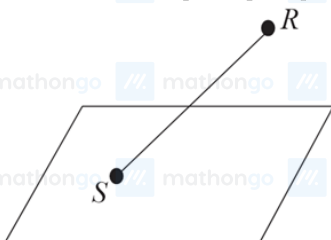
Given, R lies on plane $x - y + z = 4$

$$\therefore 3\lambda + 2 - 4\lambda + 1 + 2\lambda + 2 = 4$$

$$\Rightarrow \lambda = -1$$

$$\therefore R(-1, -5, 0)$$

Now let line SR be: $\frac{x+1}{2} = \frac{y+5}{2} = \frac{z}{1} = k$ as it is parallel to $\frac{x-7}{2} = \frac{y+3}{2} = \frac{z-2}{1}$ and passing through $(-1, -5, 0)$



So, the point $S : (2k - 1, 2k - 5, k)$

Now S lies on plane: $x + 2y + 3z = 0$

$$\Rightarrow (2k - 1) + (4k - 10) + 3k = 0$$

$$\Rightarrow 9k - 9 = 0 \Rightarrow k = 1$$

Hence, $S(1, -3, 1)$

So, distance $SR = \sqrt{4 + 4 + 1} = 3$

25. (11)

The given equations are,

$$L : \frac{x-0}{1} = \frac{y-1}{2} = \frac{z-3}{\lambda} \text{ \& } P : x + 2y + 3z = 4$$

So, Vector parallel to line: $\vec{b} = \langle 1, 2, \lambda \rangle$

Normal vector to plane $P : \langle 1, 2, 3 \rangle = \vec{n}$

Angle between plane & line is θ . Then angle between the line and normal to the plane will be $90^\circ - \theta$.

$$\text{Then, } \Rightarrow \cos(90^\circ - \theta) = \frac{(\vec{i} + 2\vec{j} + \lambda\vec{k}) \cdot (\vec{i} + 2\vec{j} + 3\vec{k})}{\sqrt{1^2 + 2^2 + \lambda^2} \cdot \sqrt{1^2 + 2^2 + 3^2}}$$

$$\Rightarrow \frac{3}{\sqrt{14}} = \frac{1 + 4 + 3\lambda}{\sqrt{\lambda^2 + 5} \sqrt{14}} \Rightarrow \lambda = \frac{2}{3}$$

$$L_1 = \frac{x-0}{3} = \frac{y-1}{6} = \frac{z-3}{2} = \mu$$

$$(x, y, z) \equiv (3\mu, 6\mu + 1, 2\mu + 3)$$

But this point lies on the plane :

$$(3\mu + 12\mu + 2 + 6\mu + 9) = 4$$

$$\Rightarrow \mu = \frac{-1}{3}$$

$$\text{Hence, } \alpha = 3\mu = -1, \beta = 6\mu + 1 = -1, \gamma = 2\mu + 3 = \frac{7}{3}$$

$$\text{Now, } \alpha + 2\beta + 6\gamma = 11$$

Hence this is the required answer.

26. (4)

Given,

The lines $L_1 : \frac{x+5}{3} = \frac{y+4}{1} = \frac{z-\alpha}{-2}$ and $L_2 : 3x + 2y + z - 2 = 0 = x - 3y + 2z - 13$ be coplanar,

Now direction ratio of line L_2 will be,

$$\vec{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 2 & 1 \\ 1 & -3 & 2 \end{vmatrix} = \hat{i}(7) - \hat{j}(5) + \hat{k}(-11)$$

Let R lies on L_2 and let $z = 0$

$$\therefore 3x + 2y = 2 \text{ and } x - 3y = 13$$

Now solving above two equation we get,

$$11y = -39 + 2 = -37 \Rightarrow y = \frac{-37}{11}$$

$$\text{And } x = 13 + 3y \Rightarrow x = 13 - \frac{37 \times 3}{11}$$

$$\Rightarrow x = \frac{143 - 111}{11} = \frac{32}{11}$$

Hence, the point $R\left(\frac{32}{11}, \frac{-37}{11}, 0\right)$

Also L_1, L_2 and given point on line L_1 to R are coplanar,

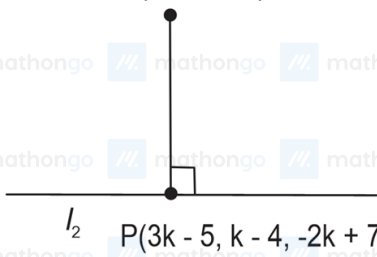
$$\text{So, by coplanar condition we get, } \begin{vmatrix} -5 - \frac{32}{11} & -4 + \frac{37}{11} & \alpha \\ 3 & 1 & -2 \\ 7 & -5 & -11 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} \frac{-87}{11} & \frac{-7}{11} & \alpha \\ 3 & 1 & -2 \\ 7 & -5 & -11 \end{vmatrix} = 0$$

$$\Rightarrow \alpha = 7$$

Now let, $P(3k - 5, k - 4, -2k + 7)$ is nearest to point $(-4, -3, 2)$

$Q(-4, -3, 2)$



So, by perpendicular condition we get,

$$\vec{QP} \cdot \vec{n}_1 = 0$$

$$\Rightarrow ((3k-1)\hat{i} + (k-1)\hat{j} + (-2k+5)\hat{k}) \cdot (3\hat{i} + \hat{j} - 2\hat{k}) = 0$$

$$\Rightarrow 9k - 3 + k - 1 + 4k - 10 = 0$$

$$\Rightarrow 14k = 14 \Rightarrow k = 1$$

$$\therefore P(-2, -3, 5) \equiv (a, b, c)$$

$$\text{Hence, } |a| + |b| + |c| = 2 + 3 + 5 = 10$$

27. (3)

We know that if a plane passes through the line of intersection of planes P_1 and P_2 then the family of planes is given by $P_1 + \lambda P_2$ where $\lambda \in R$.

Equation of plane after rotation :

$$(4x - y + z - 10) + \lambda(x + y - z - 4) = 0$$

$$\Rightarrow (4 + \lambda)x + y(\lambda - 1) + z(1 - \lambda) - 4\lambda - 10 = 0$$

Now angle between the new plane and $4x - y + z - 10 = 0$ is 90° .

$$\Rightarrow \vec{n}_1 \cdot \vec{n}_2 = 0$$

$$\Rightarrow (4 + \lambda)4 + (\lambda - 1)(-1) + (1 - \lambda)1 = 0$$

$$\Rightarrow 16 + 4\lambda - \lambda + 1 + 1 - \lambda = 0$$

$$\Rightarrow 2\lambda = -18$$

$$\Rightarrow \lambda = -9$$

$$\therefore \text{Equation of plane : } -5x - 10y + 10z + 26 = 0$$

Distance of plane from $(2, 3, -4)$

$$= \left| \frac{-10 - 30 - 40 + 26}{\sqrt{25 + 100 + 100}} \right| = \frac{54}{15} = \alpha$$

$$35\alpha = 35 \cdot \frac{54}{15} = 7 \times \frac{54}{3} = 7 \times 18 = 126$$

Hence this is the required option.

28. (3)

Let the equation of plane passing through the intersection of two given planes:

$$x + 2y + az - 2 + \lambda(x - y + z - 3) = 0$$

$$\Rightarrow x(\lambda + 1) + y(2 - \lambda) + z(a + \lambda) - 2 - 3\lambda = 0$$

This is same as $5x - 11y + bz = 6a - 1$

$$\frac{\lambda + 1}{5} = \frac{2 - \lambda}{-11} = \frac{a + \lambda}{b} = \frac{2 - 3\lambda}{6a - 1}$$

$$\Rightarrow -11\lambda - 11 = 10 - 5\lambda$$

$$\Rightarrow 6\lambda = -21 \Rightarrow \lambda = -\frac{7}{2}$$

$$\frac{2 - \lambda}{-11} = \frac{2 + 3\lambda}{6a - 1} \Rightarrow \frac{2 + \frac{7}{2}}{-11} = \frac{2 - \frac{21}{2}}{6a - 1} \Rightarrow -\frac{1}{2} = \frac{-17}{2(6a - 1)} \Rightarrow 6a - 1 = 17 \Rightarrow a = 3$$

$$\frac{2 - \lambda}{-11} = \frac{a + \lambda}{b} \Rightarrow -\frac{1}{2} = \frac{3 - \frac{7}{2}}{b}$$

$$\Rightarrow -\frac{b}{2} = -\frac{1}{2} \Rightarrow b = 1$$

$$\therefore \text{point } (a, -c, c) \equiv (3, -c, c)$$

$$\text{Distance} = \frac{2}{\sqrt{a}} = \frac{2}{\sqrt{3}}$$

$$\text{Plane: } 5x - 11y + z = 17$$

The distance of the plane from $(3, -c, c)$ is

$$\left| \frac{15 + 11c + c - 17}{\sqrt{147}} \right| = \frac{2}{\sqrt{3}}$$

$$\Rightarrow |12c - 2| = 14$$

$$\Rightarrow c = -1, \frac{4}{3}$$

$$c = -1 \quad (c \in \mathbb{Z})$$

$$\therefore \frac{a+b}{c} = \frac{3+1}{-1} = -4$$

Hence this is the required option.

29. (3)

The direction ratio of line parallel to the line of the shortest distance between the lines $\vec{r} = (\hat{i} - \hat{j}) + \lambda(2\hat{i} + \hat{k})$ and $\vec{r} = (2\hat{i} - \hat{j}) + \mu(\hat{i} - \hat{j} + \hat{k})$ is given by,

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 0 & 1 \\ 1 & -1 & 1 \end{vmatrix} = \hat{i} - \hat{j} - 2\hat{k}$$

Now given line passing through $A(-1, 2, 3)$

And having direction ratio $(1, -1, -2)$,

So, the equation is given by,

$$\frac{x+1}{1} = \frac{y-2}{-1} = \frac{z-3}{-2} = \lambda$$

Let the line intersect the plane at P

$$P \equiv (\lambda - 1, -\lambda + 2, -2\lambda + 3)$$

P lies on the plane $x - 2y + 3z = 10$

$$\therefore (\lambda - 1) - 2(-\lambda + 2) + 3(-2\lambda + 3) = 10 \\ \Rightarrow \lambda = -2$$

Hence, point $P \equiv (-3, 4, 7)$

So, the distance of point $P(-3, 4, 7)$ from point $A(-1, 2, 3)$ will be,

$$AP = \sqrt{4 + 4 + 16} = 2\sqrt{6}$$

30. (15)

The image of the point $(\frac{5}{3}, \frac{5}{3}, \frac{8}{3})$ w.r.t plane will be,

$$\frac{x - \frac{5}{3}}{1} = \frac{y - \frac{5}{3}}{-2} = \frac{z - \frac{8}{3}}{1} = \frac{-2(1 \times \frac{5}{3} - 2 \times \frac{5}{3} + \frac{8}{3} - 2)}{1^2 + 2^2 + 1^2}$$

$$\Rightarrow \frac{x - \frac{5}{3}}{1} = \frac{y - \frac{5}{3}}{-2} = \frac{z - \frac{8}{3}}{1} = \frac{-2(-1)}{6}$$

$$\Rightarrow P(x, y, z) = (2, 1, 3)$$

But $PQ = 13$

$$4^2 + 3^2 + (\alpha - 3)^2 = 169$$

$$|\alpha - 3| = 12$$

$$\alpha = 15 \text{ Since } (\alpha > 0)$$

Hence this is the required answer.

31. (1)

Let $A(0, -1, 2)$ & $B(-1, 2, 1)$ and given plane is parallel to line passing through $(5, 1, -7)$ & $(1, -1, -1)$

So, normal vector to the plane is given by,

$$\vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 0 & 2 \\ 1 & -5 & -1 \end{vmatrix} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 3 & -1 \\ -4 & -2 & 6 \end{vmatrix}$$

$$\Rightarrow \vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 3 & -1 \\ -4 & -2 & 6 \end{vmatrix} = \hat{i}(16) - \hat{j}(-10) + \hat{k}(14)$$

$$\Rightarrow \vec{n} = 16\hat{i} + 10\hat{j} + 14\hat{k}$$

$\therefore$ Equation of plane

$$16(x - 0) + 10(y + 1) + 14(z - 2) = 0$$

$$\Rightarrow 8x + 5y + 7z = 9$$

Now satisfying the points from option in the plane we get,

$(-2, 5, 0)$ only point which satisfy the plane.

32. (2)

We know that two lines $\frac{x-x_1}{a_1} = \frac{y-y_1}{b_1} = \frac{z-z_1}{c_1}$ and $\frac{x-x_2}{a_2} = \frac{y-y_2}{b_2} = \frac{z-z_2}{c_2}$ are coplanar if $\begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0$

Let us verify this option.

The given lines are $\frac{x+3}{-3} = \frac{y-1}{1} = \frac{z-5}{5}$ and $\frac{x+1}{-1} = \frac{y-2}{2} = \frac{z-5}{5}$.

$$= \begin{vmatrix} -1-(-3) & 2-1 & 5-5 \\ -3 & 1 & 5 \\ -1 & 2 & 5 \end{vmatrix}$$

$$= \begin{vmatrix} 2 & 1 & 0 \\ -3 & 1 & 5 \\ -1 & 2 & 5 \end{vmatrix}$$

$$= 2(5-10) - (1)(-15 - (-5)) + 0$$

$$= 2(-5) + (10) = 0$$

Therefore, this is the correct option.

33. (4)

Given,

N be the foot of perpendicular from the point $P(1, -2, 3)$ on the line passing through the points $(4, 5, 8)$ and $(1, -7, 5)$.

Now finding direction ratio of line L we get,

$$(4-1, 5+7, 8-5) \equiv (3, 12, 3) \equiv (1, 4, 1)$$

Hence, the equation of Line will be,

$$L : \frac{x-1}{1} = \frac{y+2}{4} = \frac{z-3}{1} = r$$

Now let point $N \equiv (r+1, 4r-2, r+3)$

And given $P \equiv (1, -2, 3)$

So, direction ratio of $PN(r, 4r-5, r+2)$

Since, $PN \perp L$ so by perpendicular condition we get,

$$r + 4(4r-5) + (r+2) = 0$$

$$\Rightarrow r = 1$$

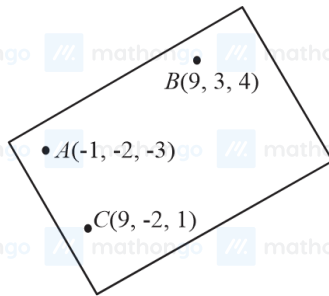
Hence, $N \equiv (2, -3, 6)$

Now finding, distance of $N(2, -3, 6)$ from plane $2x - 2y + z + 5 = 0$ we get,

$$\text{Distance} = \left| \frac{4+6+6+5}{\sqrt{4+4+1}} \right| = 7$$

34. (1)

Given that three points lie on a plane,



$$\vec{AC} = 10\hat{i} + 4\hat{k}$$

$$\vec{AB} = 10\hat{i} + 5\hat{j} + 7\hat{k}$$

$$\vec{AC} \times \vec{AB} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 10 & 0 & 4 \\ 10 & 5 & 7 \end{vmatrix}$$

$$\Rightarrow \vec{AC} \times \vec{AB} = -20\hat{i} - 30\hat{j} + 50\hat{k}$$

$$\Rightarrow \vec{AC} \times \vec{AB} = 2\hat{i} + 3\hat{j} - 5\hat{k}$$

Equation of plane will be,

$$\Rightarrow 2x + 3y - 5z = d$$

And the plane passing through, $(-1, -2, -3)$

$$\Rightarrow -2 - 6 + 15 = d$$

$$\Rightarrow d = 7$$

$$\therefore 2x + 3y - 5z = 7$$

Now finding the foot of perpendicular we get,

$$\frac{x-3}{-2} = \frac{y+2}{3} = \frac{z+9}{-5} = -\left(\frac{38}{38}\right)$$

$$x = 1, y = -5, z = -4$$

$$Q(1, -5, -4)$$

$$\text{Distance from origin} = \sqrt{1 + 25 + 16} = \sqrt{42}$$

Hence this is the required option.

35. (1)

The given lines are $\frac{x-\lambda}{0} = \frac{y-3}{4} = \frac{z+6}{1}$ and $\frac{x+\lambda}{3} = \frac{y}{-4} = \frac{z-6}{0}$

We know that the shortest distance between two lines is $d = \left| \frac{(\vec{a}_2 - \vec{a}_1) \cdot (\vec{n}_1 \times \vec{n}_2)}{|\vec{n}_1 \times \vec{n}_2|} \right|$ & $d = 13$ (given)

$$\Rightarrow \vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 4 & 1 \\ 3 & -4 & 0 \end{vmatrix} = 4\hat{i} + 3\hat{j} - 12\hat{k}$$

Now using the above formula of distance we get,

$$\Rightarrow \left| \frac{(2\lambda\hat{i} + 3\hat{j} - 12\hat{k}) \cdot (4\hat{i} + 3\hat{j} - 12\hat{k})}{\sqrt{16+9+144}} \right| = 13$$

$$\Rightarrow \left| \frac{(2\lambda\hat{i} + 3\hat{j} - 12\hat{k}) \cdot (4\hat{i} + 3\hat{j} - 12\hat{k})}{13} \right| = 13$$

$$\Rightarrow |8\lambda + 9 + 144| = 169$$

$$\Rightarrow |8\lambda + 153| = 169$$

$$\Rightarrow 8\lambda = \pm 169 - 153$$

$$\Rightarrow \lambda = \frac{16}{8}, \frac{-322}{8}$$

$$\Rightarrow 8 \left| \sum_{\lambda \in S} \lambda \right| = 8 \left| \frac{16}{8} - \frac{322}{8} \right| = 306$$

Hence this is the correct option.

36. (5)

Given,

The line $x = y = z$ intersects the line $x \sin A + y \sin B + z \sin C - 18 = 0 = x \sin 2A + y \sin 2B + z \sin 2C - 9$

So, let $x = y = z = k$

Now putting the value in $x \sin A + y \sin B + z \sin C = 18$ we get,

$$k(\sin A + \sin B + \sin C) = 18$$

Now we know that, if A, B & C are angles of triangle then $\sin A + \sin B + \sin C = 4 \cos \frac{A}{2} \cdot \cos \frac{B}{2} \cdot \cos \frac{C}{2}$

$$\Rightarrow k \left(4 \cos \frac{A}{2} \cdot \cos \frac{B}{2} \cdot \cos \frac{C}{2} \right) = 18 \dots (i)$$

$$\text{Also } k(\sin 2A + \sin 2B + \sin 2C) = 9$$

$$\text{And similarly } \sin 2A + \sin 2B + \sin 2C = 4 \sin A \cdot \sin B \cdot \sin C$$

$$\Rightarrow k(4 \sin A \cdot \sin B \cdot \sin C) = 9 \dots (ii)$$

Now dividing equation (i) & (ii) we get,

$$8 \sin \frac{A}{2} \cdot \sin \frac{B}{2} \cdot \sin \frac{C}{2} = \frac{9}{18}$$

$$\Rightarrow 80 \sin \frac{A}{2} \cdot \sin \frac{B}{2} \cdot \sin \frac{C}{2} = 5$$

37. (26)

Let the Plane containing the line $2x + y - 3z - 3 = 0 = 5x - 3y + 4z + 9$ is given by,

$$\Rightarrow 2x + y - z - 3 + \lambda(5x - 3y + 4z + 9) = 0$$

$$\Rightarrow x(2 + 5\lambda) + y(1 - 3\lambda) + z(4\lambda - 1) + 9\lambda - 3 = 0$$

Also, this plane is parallel to the line

$$\frac{x+2}{2} = \frac{3-y}{-4} = \frac{z-7}{5}$$

$$\Rightarrow \frac{x+2}{2} = \frac{y-3}{4} = \frac{z-7}{5}$$

Now normal of the plane will be perpendicular to the line, so using perpendicular condition we get,

$$\therefore (2 + 5\lambda)(2) + (1 - 3\lambda)(4) + (4\lambda - 1)5 = 0$$

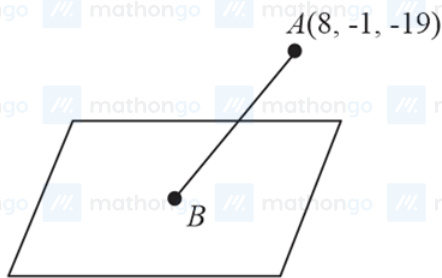
$$\Rightarrow 4 + 10\lambda + 4 - 12\lambda + 20\lambda - 5 = 0$$

$$\Rightarrow 18\lambda = -3 \Rightarrow \lambda = -\frac{1}{6}$$

So, equation of the Plane will be,

$$P: \frac{7}{6}x + \frac{3}{2}y - \frac{5}{3}z - \frac{9}{2} = 0$$

$$\Rightarrow 7x + 9y - 10z - 27 = 0$$



Now to find AB,

$$\text{Let } \frac{x-8}{-3} = \frac{y+1}{4} = \frac{z+19}{12} = k$$

$$\text{So, the point } B = (-3k + 8, 4k - 1, 12k - 19)$$

And B lies on plane P.

$$7(-3k + 8) + 9(4k - 1) - 10(12k - 19) = 27$$

$$\Rightarrow -21k + 56 + 36k - 9 - 120k + 190 = 27$$

$$\Rightarrow -105k = -210$$

$$\Rightarrow k = 2$$

$$\therefore B = (2, 7, 5)$$

$$\text{Hence, Distance is } AB = \sqrt{36 + 64 + 576}$$

$$= \sqrt{676} = 26$$

Hence this is the required answer.